Proof: Binomial identities

Jessica Taberner

Summary

Proof of binomial identities.

Before reading this proofsheet, it is recommended that you read Guide: Introduction to factorials, permutations, and combinations.

Proof

Remember from Guide: Introduction to factorials, permutations, and combinations that a **binomial coefficient** computes the number of ways of selecting k things from a collection of n things, **where order doesn't matter**.

Definition of combinations

The **combination formula** or **binomial coefficient formula** calculates the number of ways to do this, you can write this in two ways:

$$C(n,k) = \binom{n}{k} = \frac{n!}{(n-k)!k!}$$

$$C(n,k) = \binom{n}{k} = \frac{n \cdot (n-1) \cdots (n-k+1)}{k \cdot (k-1) \cdots 2 \cdot 1}$$

You would read this as "n choose k".
To use this formula n and k must be integers, with $k \leq n$. If $k > n$ then there is no way to select k things from a collection of n things, so $\binom{n}{k} = 0$.
If $n = k$ or $k = 0$ then $\binom{n}{k} = 1$ as there is only one way to pick nothing or everything from a collection of things.

Symmetry identity

> **i** Symmetry identity
>
> $$\binom{n}{k} = \binom{n}{n-k}$$

You could prove this identity many ways. You'll see 2 different ways to approach this in this sheet, both of which are interesting and valid ways of approaching the proof.

Algebraic method

First, an algebraic method: begin by using the above formula to rewrite the left hand side.

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

Then you can notice that $k = n - (n - k)$, hence:

$$\frac{n!}{(n-k)!k!} = \frac{n!}{(n-k)!(n-(n-k))!} = \binom{n}{n-k}$$

So $\binom{n}{k} = \binom{n}{n-k}$ as required!

Counting method

Another way of approaching this is by a counting argument.

$\binom{n}{k}$ is the number of ways of selecting k things from a collection of n things. This is equivalent to the number of ways of not selecting the remaining $n - k$ things.

Imagine having 2 boxes, a 'pick' box and a 'don't pick' box. For every way you can put k things in the 'pick' box, there's a way to put $n - k$ things in the 'don't pick' box. If you imagine relabeling these boxes you can see that $\binom{n}{k}$ must equal $\binom{n}{n-k}$.

Pascal's identity

> **Pascal's identity**
>
> $$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Again, you could prove this identity many ways. You'll see 2 different ways to approach this in this sheet.

Algebraic method

First, an algebraic method: begin by using the formula to rewrite the right hand side.

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{(n-1)!}{(n-k)!(k-1)!} + \frac{(n-1)!}{(n-k-1)!k!}$$

In order to make the denominators equal, you can multiply the left fraction by k and the right fraction by $n-k$.

$$\frac{k(n-1)!}{(n-k)!k!} + \frac{(n-k)(n-1)!}{(n-k)!k!} = \frac{(k+n-k)(n-1)!}{(n-k)!k!}$$

Simplifying and rewriting the right hand side you can get $\binom{n}{k}$ as required.

$$\frac{(k+n-k)(n-1)!}{(n-k)!k!} = \frac{n(n-1)!}{(n-k)!k!} = \frac{n!}{(n-k)!k!} = \binom{n}{k}$$

Counting method

Alternatively, you could also approach this with a counting argument.

Imagine having a collection with n items, and picking one. Let's call this 'star'. If you want to pick k elements from the collection, it either includes 'star' or it doesn't. If it does include 'star' then you need to pick $k-1$ elements from $n-1$ elements, there are $\binom{n-1}{k-1}$ ways to do this. If it does not include 'star' then you need to pick k elements from $n-1$ elements, there are $\binom{n-1}{k}$ ways to do this. Adding these together gives the result.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Further reading

For more on this topic, please go to Guide: Introduction to factorials, permutations, and combinations.

Version history

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